Geochimica et Cosmochimica Acta 406 (2025) 1–4

Contents lists available at ScienceDirect

Geochimica et Cosmochimica Acta

journal homepage: www.elsevier.com/locate/gca

Editorial

Hydrothermal geochemistry: a tribute to Terry M. Seward☆

Terry Maxwell Seward (1940–2022) was a pioneering figure in the field of hydrothermal geochemistry (Fig. 1). His extensive research greatly impacted the scientific community, particularly in advancing our understanding of the transport of ore-forming metals in hydrothermal fluids. Through his experimental work and theoretical developments, Terry provided novel insights into the geochemistry of aqueous and volatile species in hydrothermal fluids, shaping modern thermodynamic models of fluid-rock interactions, organic compounds, and the complex chemistry of volcanic fumaroles. He also advanced ideas on stable isotope fractionation in hydrothermal systems, expanding our capacity to trace fluid sources and processes in Earth's crust. Beyond his scientific impact, Terry's enthusiasm for minerals shaped both his personal and professional life. Terry was a generous mentor, collaborator, and teacher who inspired many geochemists.

This special volume, *A Tribute to Terry Seward*, honors the outstanding legacy of a pioneering scientist in hydrothermal geochemistry. The collection features a broad range of contributions that reflect Terry's deep influence across the field, including work on metal complexation, solubility, isotope fractionation, and fluid-mineral interactions. Topics span from the behavior of antimony and gold in hydrothermal systems to the thermodynamics of rare earth elements and the role of fluids in mineralization. Many of these papers are authored by Terry's close colleagues, former students, and postdocs, underscoring his lasting impact as both a mentor and scientific leader.

1. Journey of a hydrothermal geochemistry pioneer

Terry Seward was born in 1940 in Toronto (Canada). Throughout his life, he was known for his friendliness, creativity, and art of storytelling, traits that characterized his everyday interactions with colleagues and students.

Terry began his education in geology and chemistry at McMaster University (Canada), where he graduated with a Bachelor of Science in 1965. He and his wife, Diane Seward, moved to Manchester (United Kingdom) where Terry earned his Ph.D. in 1969 under the supervision of William Fyfe. After graduation, Terry joined the Department of Scientific and Industrial Research in Lower Hutt (New Zealand), where he blended experimental and field studies that would later define his career. Under Terry's influence (1970–1989), the facility became a center for pioneering research, providing critical data that enabled quantitative assessments of hydrothermal fluid chemistry and ore deposit formation.

Between 1976 and 1978, Terry spent time at the University of Karlsruhe (Germany) as an Alexander von Humboldt Fellow at the Institute for Physical Chemistry. During this period, he conducted research in physical and inorganic chemistry, focusing on hydrogen and water at high-temperature and high-pressure conditions. In 1981, Terry spent a year on sabbatical at Pennsylvania State University (USA) working on the geochemistry of CO_2 and emissions from geothermal fields, a subject that later became of greater interest to the scientific community.

In 1988–1989, Terry left New Zealand to join the Swiss Federal Institute of Technology in Zurich (ETH-Zurich, Switzerland) as a guest professor, before being appointed as a full professor in 1990. There, he established a new laboratory dedicated to high-temperature geochemistry research, leading projects on the thermodynamics and kinetics of inorganic and organic compounds in hydrothermal fluids and on the composition of volcanic fumarolic gases.

In 2009, Terry left ETH-Zurich to join Stanford University (USA) as a Cox Visiting Professor. In 2012, he spent a year as a visiting scientist at Tongji University in Shanghai (China). He then returned to New Zealand in 2013 to take up a position as Professor of Geochemistry at the Victoria University of Wellington.

2. Hydrothermal fluids

Terry's primary research interest was hydrothermal fluid geochemistry. His work bridged fundamental experimental, theoretical chemistry and applied geosciences, producing foundational research that remains central to modern geochemistry.

One of Terry's most important contributions was the experimental measurement of metal solubility under hydrothermal conditions. Working primarily at elevated pressures and temperatures representative of crustal to deep Earth environments, he systematically investigated the solubility and complexation of key ore-forming metals such as, copper, silver, gold, and zinc (Benning and Seward, 1996; Mountain and Seward, 2003; Seward, 1973, 1976; Stefánsson and Seward, 2004; Tagirov and Seward, 2010). These studies elucidated the critical role of chloride and sulfide ligands in stabilizing metal complexes in solution, thereby controlling the transport and deposition of metals in hydrothermal ore systems. His thermodynamic data remain key for understanding and quantifying ore-formation processes.

To unravel the speciation of aqueous complexes *in situ*, Terry developed and applied UV–Visible (UV–Vis) absorption spectroscopy under high-temperature conditions. By pioneering custom-designed optical cells, he directly monitored metal–ligand equilibria in aqueous

☆ This article is part of a special issue entitled: 'A Tribute to Terry Seward' published in Geochimica et Cosmochimica Acta.

https://doi.org/10.1016/j.gca.2025.08.033

Available online 23 August 2025
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Fig. 1. Upper left: Terry at the Department of Scientific and Industrial Research (New Zealand) during the 1970s. Upper right: Terry collecting volcanic gas samples at Vulcano (Italy) in the 1990s. Below: Terry in his garden in Wellington (New Zealand).

solutions with various complexing ligands at temperatures exceeding 300 °C. His early work on lead-chloride complexation remains a defining example of such work (Seward, 1984), which guided subsequent studies including those sulfide (Suleimenov and Seward, 1997), molybdate (Minubayeva and Seward, 2010), arsenite (Zakaznova-Herzog et al., 2006), palladium (Boily and Seward, 2005b), and iron (Stefánsson et al., 2019). This body of work provided key thermodynamic data — such as stability constants and absorption coefficients — for species that are otherwise difficult to isolate or characterize, offering a robust basis for geochemical modeling.

Expanding on his UV–Vis work, Terry employed Extended X-ray Absorption Fine Structure (EXAFS) spectroscopy to probe the local structure and coordination environment of dissolved metal complexes in hydrothermal fluids, becoming a pioneer for this kind of work on hydrothermal solutions. His work using synchrotron radiation to investigate speciation of metals, such as silver and iodide (Charnock et al., 1999; Seward et al., 1996; Seward et al., 1999), under realistic geological conditions offered fundamental and atomic-level insight into metal–ligand interaction and transport mechanisms. These studies demonstrated the capacity of EXAFS to resolve hydration numbers, ligand identity, and bond lengths, deepening the understanding of fluid chemistry in ore-forming systems. Terry also applied other types of *in situ* measurements to understand aqueous speciation, for example Raman spectroscopy to quantify the complex nature of sulfur species in aqueous solutions (Schmidt and Seward, 2017).

In addition to metal speciation, Terry and his colleagues investigated the behavior and stability of organic compounds in hydrothermal systems (Cox and Seward, 2007b; Boily and Seward, 2005a; Cox and Seward, 2007a; Sajkowski et al., 2022). This research included both experimental work and thermodynamic modeling, driven in part by the recognition that organic ligands could influence metal mobility. He showed that organic molecules can remain stable at high temperatures and participate actively in ore deposition processes, especially in sediment-hosted and volcanic hydrothermal systems.

Terry made key advances in understanding hydrothermal fluids by integrating molecular dynamics and quantum chemical simulations with experimental findings (Boily and Seward, 2005b; Driesner et al., 1998; Lemke and Seward, 2008; Stefánsson et al., 2019; Suleimenov et al., 2003). His work explored ion hydration and pairing in aqueous solutions from ambient to supercritical conditions, revealing how ion solvation changes with temperature and concentration and the nature of water as a solvent (Driesner et al., 1998). He also studied isotope fractionation by

modeling hydration complexes of alkali and halide ions, providing atomic-level insights into the isotopic composition of natural waters (Driesner et al., 2000).

He also applied a broad range molecular simulations together with mass spectrometric techniques to probe molecular clusters in high temperature gases. A key feature of this effort was the construction of the first pulsed ion-mass spectrometer at ETH Zurich, an experimental set-up specifically designed to explore the hydration behavior of molecular ions in high-temperature aqueous vapors. This mass spectrometric work advanced knowledge of ion hydration and speciation, proving crucial for understanding vapor-driven transport (Likholyot et al., 2007; Lemke and Seward, 2018).

Terry's research even encompassed the geochemistry of volcanic gases, spanning metals, organics and major gases like sulfur. And he visited many volcanoes. At White Island (New Zealand), Terry contributed to studies on magma mixing, degassing, and late sulfide saturation during eruptive sequences, elucidating metal behavior in evolving volcanic systems (Mandon et al., 2021). At Vulcano (Italy), he investigated diffuse emissions of organic trace gases and halocarbons, revealing diverse heteroatomic organic compounds in volcanic gases and their atmospheric implications (Schwandner et al., 2013). At Kudriavy volcano in the Kurile Islands (Russia) Terry studied volatile trace-metal transport (Wahrenberger et al., 2002), and at Ol Doinyo Lengai (Tanzania) he investigated trace-element distribution between fumarole condensates and natrocarbonatite lavas (Teague et al., 2008). His multidisciplinary approach combining field observations, geochemical analyses, and thermodynamic modelling, advanced our understanding of volatile-metal–organic interactions in volcanic environments.

3. A passion for minerals

Terry had not only a keen interest for hydrothermal fluids but also was well known among colleagues and friends for his passion for minerals. From a young age, he was fascinated by the natural beauty and diversity of mineral specimens. He joined the Junior Walker Mineralogical Club in Toronto where his father, Max Seward, led many field trips. This was an enthusiasm that he carried throughout his life. Over the years, Terry assembled an impressive collection of rare and unusual minerals, with a particular interest in arsenates, sulfides, and hydrothermal phases.

One notable example of his mineralogical legacy is the mineral sewardite — a rare calcium-iron arsenate [$CaFe_2^{3+}(AsO_4)_2(OH)_2$] — named in Terry's honor (Roberts et al., 2002). Terry discovered the original specimen during a trip to Namibia, a country he visited regularly for both research and mineral collecting. Recognizing the uniqueness of the specimen, Terry encouraged colleagues to investigate the crystal composition further, ultimately leading to formal identification and naming. This deep appreciation for minerals helped Terry cultivate a lifelong scientific curiosity that shaped his career in geochemistry.

4. Honours and legacy

Throughout his distinguished career, Terry received numerous accolades in recognition of his pioneering contributions to geochemistry. He notably became a Fellow of the Geochemical Society in 2000, and in 2005 he received the Robert Wilhelm Bunsen Medal of the European Geosciences Union. His work remains highly respected and continues to be widely cited, reflecting its lasting value to both colleagues and students. Terry was also a committed mentor and a leader within the geochemical community. He served as an editor for *Geochimica et Cosmochimica Acta* and held prominent roles in several scientific societies, including as President of the European Association of Geochemistry.

Equally important was his dedication to mentoring early-career scientists. Many of those he taught and inspired are now leading researchers. His generosity with time, ideas, and encouragement left a profound mark on the international geochemistry community. Terry's legacy lives on in the continued development of hydrothermal geochemistry, and in the many scientists who carry forward the principles and curiosity he so strongly embodied.

Finally, Terry's influence extended well beyond the laboratory. Known for their warmth and generosity, he and his wife Diane welcomed students, postdocs, and collaborators from around the world into their home in Switzerland and New Zealand. Terry's hospitality, humor, and dedication to mentoring have left a lasting legacy in the geochemical community.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to extend our gratitude to the Executive Editors who contributed to this special issue. Jeffery Catalano, for endorsing and making this project possible, and Hailiang Dong for guiding it to completion. We thank all authors for their scientific contributions and the reviewers who provided their expertise in the evaluation process. We are grateful to Jenny Cox and Peter Tremaine for providing details about Terry's background. We also wish to express our heartfelt appreciation to Diane Seward, for her unwavering support. Diane, an accomplished geoscientist, was a constant source of strength for Terry and for his group members. Finally, we are deeply grateful to Terry for his mentorship and guidance throughout our careers. His support has played an important role in our development as geochemists.

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